Designing an Accessible and Portable Gaming Controller for Guitar-Style Rhythm Games

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Introduction

This report represents the design and development of a unique, accessible and portable gaming controller for Guitar-style rhythm games. The design of the controller is intended to mimic the intuitive use of a guitar, while in a smaller, more accessible package for all users. The primary objective of this project is to develop a functional prototype of the developed controller design, using custom and consumer electronic components, alongside a custom 3D printed housing shell. Both hardware and software components were developed with our primary objective in mind. Through this prototype we have expressed our idea for an alternative controller that can help a variety of users engage with guitar-style rhythm games in a new or more accessible manner. Our experiences, methods, and results are documented in this report.

Materials and Methods

The controller design we have developed incorporates various custom and consumer electronic components to accomplish the desired results. A custom PCB layout, designed in KiCad, combined with low-profile mechanical keyboard switches for comfort and ease of use. Additionally, the Seeed Xiao nrf52840 microcontroller which connects via a USB-C to USB-C cable allows for easy connectivity to the desired platforms. The firmware to automatically have the controller work on any device is ZMK. Done by configuring files such as: overlays, .shield, .build, and .layouts to our liking, then ran through a Github workflow zmk template. This outputs a uf2 file which has the ability to flash onto peripheral devices like our microcontroller. All of this put together with M3 screws and appropriate inserts makes for a rigid controller that can be disassembled for maintenance or case replacement.

Our shell was designed using Fusion360 and went through four iterations to get to the final prototype. The multiple attempts allowed us to experiment with different approaches, from a very large controller to a very small controller idea, before settling in the middle where we found it was comfortable and accomplished our goals for portability. The shell of the controller is 3D printed out of PLA plastic to give a rigid and stable enough body for our use-case.

Results

The following is a summary of the results of a SUS and TLX questionnaire. This questionnaire was given to users after they first interacted with our prototype controller. Our aim was to evaluate comfort, usability, portability, accessibility, and performance as referenced in B the file can be located, and the table in Appendix C, user testing results had a common problem of handling due to the shape and size of the controller.

The majority of users found the controller simple in its design and functional in its integration. The controller was seen as uncomplicated and easy to use, however some users found the shape slightly awkward when finding an initial grip position. This was followed by a TLX testing section which yielded the following results (also shown in Appendix C). An average score of 58.53 indicates a higher-than-average workload. While the controller is simple to understand, being able to perform in the task of playing a rhythm game was challenging, depending on the users' experience level. Those who had experience stated that, while unfamiliar at first, it was enjoyable after they had the chance to adjust to the smaller size and feel.

Takeaways and Recommendations

A major takeaway for the shell designing section was to always check with the 3D printer before deciding if a design was final. Our third iteration looked good as a render, but once it was sliced and prepared to print, there were several issues surrounding the printability. From how long it would take to print to the shell just being too large, there are always benefits to double checking and saving yourself another adjustment or two. In terms of hardware and electronics, it would have been a lot smarter to order our parts earlier as it would have saved a lot of stress and time. A lot of easy fixes were present during assembly and soldering, but the stress of time clouded perception. It also would have been smart to get certain preliminary pieces done ahead of time. This could have been the KiCad parts, schematics, PCB layout, getting ZMK firmware ready, leaving more time for soldering, assembly and errors. One last recommendation or takeaway would to be to consult the Teaching Assistant. Constantine is very helpful, knowledgeable, and insightful. Things like embarrassment and fear are one time but getting it over with sooner means reaching the goal faster, and again saves time and stress. He was especially helpful with ZMK debugging, but also with hardware. Recommending certain schematic libraries, websites to get parts like the microcontroller and PCBs. He was also helpful with the soldering and assembly, recommending flux and threaded inserts.

Appendix A

Key Cap: https://www.amazon.ca/Generic-Chocfox-CFX-Keycaps-Chosfox/dp/B0CP27QXG7/ref=sr_1_5?dib=eyJ2IjoiMSJ9.s3UqDrjQXWO-PjVacTRKEmifF5BlmwU2VeUj1NYA0k5GVMteIrX-ovxRynPpZNjtT4M6C044dFUxA7e-iylRlVpHDonDDJuYhzTBNIC6aT6A3ayBxQsFEWpVs1y6t_WcsMzKwJyDqWH8OgxugDY9WKZrnYjRWQrrEI1Fi6hsWmMbAEVNOiYOCIp5I1NfN7BxxGNErMbclPwjQgbz2o8kLqMQnWnnU4r6HyN0QQkvB8CnRbe7KBYTOFYNKsy3Co67jzHvo2BzVB8tkUzqPqg_I1wHWpcTy8MzBnnSn6lGf0.p2rSIqtQxgKiowosfMdyB7RO1e4_pDVWM-qDV7kOMRA&dib_tag=se&keywords=kailh%2Bchoc%2Bv1%2Blow%2Bprofile%2Bkeycaps&qid=1775613909&sr=8-5&th=1

Key Switch: https://www.amazon.ca/KAILH-Official-Chocolate-Mechanical-Keyboard/dp/B0B3MLZJ8K/ref=asc_df_B0B3MLZJ8K?mcid=52fecee5698c31c8bc430a032e739c39&tag=googleshopc0c-20&linkCode=df0&hvadid=753787984763&hvpos=&hvnetw=g&hvrand=10443908158767223747&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9000833&hvtargid=pla-2247354483984&hvocijid=10443908158767223747-B0B3MLZJ8K-&hvexpln=0&gad_source=1&th=1

Jumper Wires: https://www.amazon.ca/RGBZONE-Multicolored-Breadboard-Arduino-Raspberry/dp/B08TWSV2DY/ref=sr_1_7?crid=16NC40Q2VJLXL&dib=eyJ2IjoiMSJ9.1JTtZYzqh1JVSNxn_zOlNE4ejRk_z5Dv6xKvsANVUEKXxCnE01NWCwbBFLGUdiXMqkyv9yYj1HJK6Oog1E6ZTitVh29hoJKKdaZ1E7pSS2x0GcNV1dQR4HXzGF84oJQR7jwhv6muA25e2r2EKv6Mz9toG_yd4lALyyghdd7uCYAK39jtlWeN5nPTML4vTNeHnDnUkE33gCwykhoDbUuoYClVZ3oh27Y-rvSvG2GV1EEHJMU_zhtvB2IDPuvx9n14fdBz2OgYHgTcpQuTzNwYTZbJl0g8PkED7_QGXyjVXas.-aou8Dq3KCs8eFwocjj-Cvldhuwz3iGgiXgIVOC5oaM&dib_tag=se&keywords=jumper+wires&qid=1775614087&s=industrial&sprefix=jumper+wi%2Cindustrial%2C205&sr=1-7

Threaded Inserts: https://www.amazon.ca/Threaded-Printing-Embedment-Automotive-4-6x5-7mm/dp/B0BYVLPG4R/ref=asc_df_B0BYVLPG4R?mcid=26d114182bf6360f990d2fd05b5a25dd&tag=googleshopc0c-20&linkCode=df0&hvadid=780651366195&hvpos=&hvnetw=g&hvrand=13191125959132108238&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9000833&hvtargid=pla-2188367812775&psc=1&hvocijid=13191125959132108238-B0BYVLPG4R-&hvexpln=0&gad_source=1

Appendix B

Deliverables Github: https://github.com/boshna1/indust.-game-design-hardware-project-RythymGuitarController

Appendix C

TLX Results:

Task: Play *Clone Hero* for ~10 minutes using the custom controller.

MD	PD	TD	PF	EF	FR	R-Score
40	20	50	60	30	20	36.66
40	60	80	50	60	30	53.33
40	70	80	50	70	30	56.66
80	90	80	80	80	50	76.66
40	60	60	20	70	30	46.66
70	70	70	80	60	50	66.66
80	90	90	90	80	60	81.66
70	60	60	30	60	20	50

References

Microcontroller (Seeed Xiao): https://wiki.seeedstudio.com/XIAO_BLE/

ZMK Firmware Github (Used for Workflow and production of uf2 file): https://github.com/zmkfirmware/zmk